TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

PROJECT SYNOPSIS

Student Details

Institute/University	Pimpri Chinchwad College of Engineering (PCCOE), Pune
Programme / Batch	B.Tech Computer Science and Engineering / 2027 Batch
Student Name	Bhagawati Kadam
PRN	123B1B154

1. Case Study ID and Project Title / Topic

Case Study ID (CS1 to CS5)	CS2
Project Title / Topic	Functional Safety HARA and Safety Requirement Assistant

2. Objectives

The project reduces the manual effort of the Hazard Analysis and Risk Assessment (HARA) and of writing safety requirements for an automotive item, while keeping risk classification deterministic and every entry approved by an engineer.

- Build an assistant that helps a functional safety engineer identify malfunctioning behaviours and hazardous events for an Electric Power Steering (EPS) item, rate them for severity (S), exposure (E) and controllability (C), and derive the ASIL.
- Use RAG to retrieve similar approved hazards, the item definition and the HARA rules as evidence, with citations to item block, hazard ID and rule ID.
- Use a local LLM to draft hazard descriptions, safety goals, functional safety requirements (FSR) and technical safety requirements (TSR), which the engineer edits and approves.
- Keep the ASIL, the requirement-quality check and the traceability-gap check deterministic (Python rules), and record every decision with CSV, JSON and Markdown export.

3. Scope

The scope is intentionally limited to a controlled prototype with synthetic content, deterministic rules and human approval. It does not claim ISO 26262 compliance.

In Scope – To Be Implemented

- A synthetic EPS item definition (9 blocks, 5 functions), an approved hazard library of 14 entries and a set of 23 HARA, requirement and traceability rules as the knowledge base.
- RAG-based Q&A, similar-hazard retrieval and AI drafts for hazards, safety goals, FSR and TSR, with an automatic citation check.
- Deterministic ASIL calculation, a requirement-quality checker and a traceability matrix with gap detection (hazard → safety goal → FSR → TSR).
- Local/private processing using Ollama, Sentence Transformers and FAISS, a Streamlit interface, SQLite storage and export of the HARA worksheet and traceability matrix.

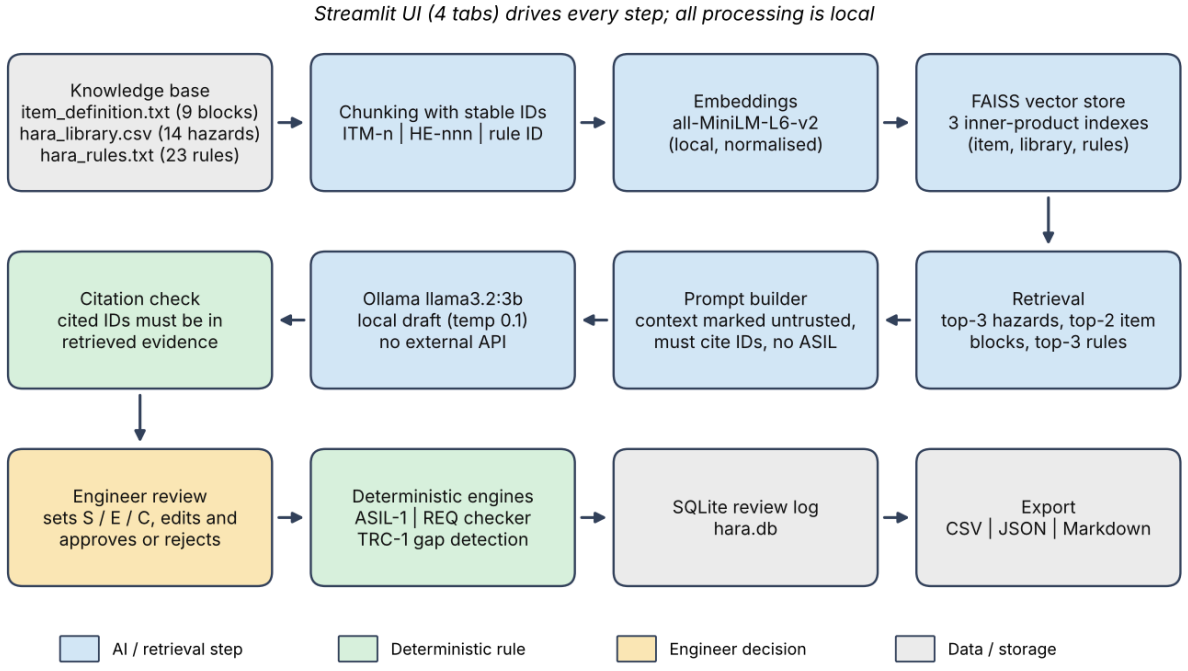
Out of Scope – Will Not Be Implemented

- Automatic approval of ratings or requirements, certification or ISO 26262 compliance claims, and use of proprietary standards or OEM data.
- Other safety analyses (FMEA, FTA, safety case), multi-item projects and production-scale deployment.

4. Proposed Solution Design

Component	Planned choice
Pre-trained model (name, version, local/private hosting)	Ollama running locally with llama3.2:3b (temperature 0.1). Used for grounded answers and for drafting hazards, safety goals and requirements. No external LLM API.
Embedding model	Sentence Transformers all-MiniLM-L6-v2 for item blocks, library hazards, rules and user queries.
Vector store (FAISS / ChromaDB)	FAISS with three local inner-product indexes: item blocks, approved hazard library and rules.
RAG pipeline and/or AI-agent design – brief description	Each item block, library hazard and rule is one chunk with a stable ID (ITM-n, HE-nnn, rule ID). For a query, the top 3 hazards, top 2 item blocks and top 3 rules are retrieved and passed to the local LLM. The prompt treats the context as untrusted, requires citing IDs, forbids the model from deciding the ASIL and asks for 'Not found in supplied context.' when evidence is missing. A citation check flags cited IDs that were not retrieved. No autonomous agent is used.
Service layer and UI (e.g., FastAPI / Flask / Streamlit)	Streamlit interface with four tabs (Item & Q&A, Hazard analysis, Safety goals & requirements, Traceability & export) and Python modules for retrieval, LLM calls, rules and storage. SQLite stores the HARA log.
Tools / deterministic rules, if applicable	asil.py (rule ASIL-1: S+E+C sum; any class 0 gives QM), reqcheck.py (requirement checks REQ-1 to REQ-5) and traceability.py (rule TRC-1). The LLM never decides these results.

Proposed System / Architecture



System architecture (also included in the technical report).

5. Input Data / Knowledge Base

Only synthetic project content is used as the knowledge base.

Data / Knowledge Sources	A synthetic item definition of an Electric Power Steering system (item_definition.txt), a synthetic approved hazard library (hara_library.csv) and a rules file (hara_rules.txt) covering ASIL determination, requirement quality, safety goals and traceability. No live or proprietary data is required.
Approximate Size and Format	item_definition.txt: 9 text blocks (ITM-1 to ITM-9). hara_library.csv: 14 approved hazardous events with function, malfunction, hazard, situation, S, E, C, ASIL and rationale. hara_rules.txt: 23 rules with IDs. HARA entries, goals and requirements are stored in local SQLite and exported as CSV, JSON and Markdown.
Preprocessing and EDA Planned	Text is split into one chunk per item block, library entry and rule, each with a stable ID, then embedded with all-MiniLM-L6-v2 and indexed with FAISS. Numerical EDA is not required because the knowledge source is short engineering text; the library ASIL values are checked against the rule for consistency.

6. Expected Outcomes and Evaluation Plan

Expected Outputs – aligned to the case study	A HARA worksheet with S/E/C and rule-computed ASIL per hazardous event; safety goals with ASIL, safe state and FTTI; FSR and TSR that pass a quality check; a traceability matrix with a list of gaps; cited AI drafts; and an exportable record of every engineer decision. The aim is to reduce manual effort while the ASIL and checks stay deterministic and decisions stay human-controlled.
Evaluation Method – test question set / ground truth / metrics / validation approach	Scripted evaluation (run_eval.py): (1) ASIL rule against reference S/E/C combinations; (2) consistency of the 14 library ASIL values with the rule; (3) requirement checker on a set of compliant and defective requirements; (4) traceability check on injected gaps, measuring detected gaps and false alarms; (5) retrieval hit rate in the top 3 for a set of test questions with an expected hazard ID; (6) citation validity and checker pass rate of AI-drafted safety goals and FSRs. Sample AI outputs are also reviewed manually. Results are saved in Evaluation_Results/.
Responsible AI Measures – grounding and citations / privacy / security / human oversight	Grounding: answers use retrieved evidence and cite IDs, and a citation check flags unsupported IDs. Privacy: the LLM, embeddings and vector store run locally. Security: retrieved text is treated as untrusted in the prompt; nothing is approved or executed automatically. Human oversight: an engineer approves or rejects every hazardous event, safety goal and requirement. Only synthetic data is used and ratings are illustrative.
MLOps / Deployment Elements Planned – versioning / Docker / tracking	Git/GitHub for source versioning, a Python virtual environment and requirements.txt for dependencies. The prototype runs locally; Docker, CI/CD and formal experiment tracking are future improvements.

7. Planned AI-Tool Usage

AI tools are used as development assistance and as a local model inside the application; the student reviews the resulting implementation.

AI Tool / Model	Purpose	Artifact Affected
Claude (Anthropic)	Coding assistance, generation of the synthetic knowledge base and test cases, debugging, and drafting of the synopsis and technical report.	Python modules, synthetic knowledge base, evaluation cases, synopsis and technical report. All assisted content is reviewed and verified by the student.
Ollama / llama3.2:3b	Local project LLM for Q&A and for drafting hazards, safety goals and requirements.	RAG/AI modules and generated draft content.

Student Declaration

☑ I will not change the approved case-study scope, major methodology, or implementation approach without prior faculty approval.

☑ I confirm that the work submitted under this synopsis represents my own understanding and contribution, and I will be responsible for explaining every significant part of the selected case study and its implementation.

Signature:		Date:	07-10-2026
Name:	Bhagawati Kadam	PRN:	123B1B154

Faculty Approval Record

Project title	Functional Safety HARA and Safety Requirement Assistant (CS2)
Student	Bhagawati Kadam, PRN 123B1B154
Faculty / mentor name	
Decision	☐ Approved ☐ Approved with changes ☐ Not approved
Remarks	
Faculty signature and date	